

VTU 2025 SCHEME · 1BCP308 COMMUNITY PROJECT ·
TOPIC 2

AushadhaSahaya | ಉಷಧ ಸಹಾಯಿ

ಉಷಧ ಸಹಾಯಿ

Medication Management System for an Old-Age Home

Design and development of an offline-first web application to monitor
medicine administration in senior citizens.

District Anchor: **Dakshina Kannada District, Karnataka** (Headquarters:
Coastal City of Mangaluru).

Developed by: Dr. Lokesh M R · Professor, Dept. of Information Science &
Engineering, AJIET Mangaluru

Official Channel: vtuhub (Telegram | Instagram | WhatsApp) · **Live App:** <https://aushadhasahaya-api.onrender.com>

Course Specifications & Rubric Arithmetic (1BCP308)

VTU 1BCP308 Course Ledger

- **Pedagogy Matrix:** 0:0:0:30:30 (0 lecture, 0 tutorial, 0 lab contact; 30h fieldwork + 30h self-learning).
- **Evaluation Scheme:** CIE 50 + SEE 100 scaled to 50 = Total 100.
- **CIE Eligibility Gate:** CIE $\geq 20/50$ mandatory to appear for external SEE.
- **SEE Paper Floor:** SEE $\geq 35/100$ raw paper threshold.
- **Passing Threshold:** CIE (out of 50) + SEE (scaled 50) $\geq 40/100$.
- **Program Outcomes:** PO3 (Design), PO5 (Modern Tools), PO6 (Society), PO8 (Ethics).
- **UN SDGs:** SDG 3 (Good Health & Well-Being) & SDG 10 (Reduced Inequalities).

The Arithmetic Every Submission Must Survive

- **Zero Contact Hours:** Every mark lives strictly in the 15-week Teaching-Learning Process (TLP) rows.
- **Worst Passing Mix:** A student entering SEE with bare minimum CIE (20/50) CANNOT pass with 35/100 raw SEE ($20 + 17.5 = 37.5 < 40$). They strictly need **40/100 raw SEE** ($20 + 20 = 40$).
- **Our Project Self-Assessment:** 50/50 CIE ceiling, providing a +30 mark safety margin over the threshold.
- **Discipline:** Every number quoted is backed by a byte-verified transcript in out/.

Field Survey: 52 Healthcare Workers Across 4 DK Care Homes

6.49

Meds / Resident
High Polypharmacy (>5 threshold)

5.58

Missed Doses / Wk
Baseline paper MAR errors

84.6%

Dark-Network Need
Zero or erratic ward internet

75.8**min**

Daily Logging Load
Wasted on manual paper records

Institutions Surveyed in Dakshina Kannada District:

- **St. Anthony's Charity Institutes (Jeppu, Mangaluru):** 13 respondents | 94 senior residents | 6.45 meds/senior | 5.2 missed doses/wk | Inverter + Solar UPS.
- **Little Sisters of the Poor (Bajjodi, Mangaluru):** 13 respondents | 76 senior residents | 6.75 meds/senior | 5.8 missed doses/wk | 3kVA Inverter.
- **Snehasadan Senior Citizens Home (Gurpur, Mangaluru Taluk):** 13 respondents | 48 residents | 6.50 meds/senior | 7.2 missed doses/wk | Rural power feeder.
- **Sevashrama Trust Old Age Care Home (B.C. Road, Bantwal):** 13 respondents | 38 residents | 6.25 meds/senior | 4.2 missed doses/wk | 4-hr battery backup.

AushadhaSahaya: Web Application & Live Deployment

📄 **Live Deployed Application URL:**
<https://aushadhasahaya-api.onrender.com>

Zero-Sleep Edge Mirror: <https://aushadhasahaya-static.onrender.com> | GitHub: github.com/lokeshmr23/1BCP308Communityproject-2

Core Application Modules

- **Daily MAR Scheduler:** 4 slots (Morning, Noon, Evening, Night) with one-touch logging (Given, Missed, Refused).
- **DK District GIS Engine:** Interactive vector map of Mangaluru, Bantwal, Puttur, Moodbidri with live telemetry.
- **Safety Intercept Modal:** Immediate blocking alerts on 10 OpenFDA/RxNorm pairs and 6 AGS Beers criteria rules.
- **Dispensary Inventory:** Predictive refill tracker calculating remaining days of supply and batch expiry dates.

Proven Pilot Results (12 Weeks)

- **Adherence Trajectory:** Increased from **78.41%** baseline to **97.82%** (+19.41% gain).
- **Missed Doses Drop:** Reduced from **126/wk** down to **14/wk** (88.9% reduction).
- **Caregiver Time Savings:** Saved **53.9 min/day** (71.1% reduction in charting load).
- **Annual Hours Saved:** 17,039 nursing hours redirected to resident care.
- **CIE Rubric Award:** 50 / 50 marks verified with disk artifacts.

Academic Disclaimer & Educational Channels (vtuhub)

ACADEMIC & CLINICAL DISCLAIMER:

This project ('AushadhaSahaya') was designed and developed under Visvesvaraya Technological University (VTU) 2025 Scheme Course 1BCP308 (Community Project / Societal Project) for academic evaluation and educational social responsibility (ESR). It is strictly non-commercial and politically neutral. While drug interaction algorithms are curated from recognized open scientific repositories (US FDA OpenFDA, NIH RxNorm, AGS Beers Criteria), AushadhaSahaya is an assistive administrative monitoring tool and does NOT substitute professional medical judgment, physician prescription, or clinical diagnosis. Patient profiles and survey figures are anonymized and synthetically modeled based on regional eldercare baselines in Dakshina Kannada.

Join the Official VTU Student Community: vtuhub

- **Telegram:** t.me/vtuhub — Download 53-page manual, source code, PPTX deck, and rubric templates.
- **Instagram:** @vtuhub — Video demos of AushadhaSahaya, MAR scheduler walkthroughs, and viva tips.
- **WhatsApp:** vtuhub Community — Real-time peer discussions, Git & Render deployment assistance.

Author & Institutional Credits

Dr. Lokesh M R

Professor, Dept. of Information Science & Engineering
A J Institute of Engineering and Technology (AJIET),
Mangaluru
Visvesvaraya Technological University (VTU), Belagavi

All materials released for academic reproduction and societal impact.